18.06 — Linear Algebra

100 points

Unit 1 Exam — Graded

Fall 2011

Student: Amin Cheipournejad · Total: 77.5 / 100

Grade Report

Part	Score	Grader's comment
Q1 (a)	**4 / 8**	The number of rows is right — three, since the right-hand sides have three entries — and so is the range of the rank, one or two. The number of columns is not. The answer argues that A may have any number of columns, sketching a matrix with repeated columns, but a repeated column gives Ax = 0 a nonzero solution, and adding it to the solution for (0, 1, 0) gives a second one. Exactly one solution means no free variables, so the columns are independent and n = r. The only possibilities are m = 3 with r = n = 1, or m = 3 with r = n = 2; that relation is the main content of the part.
Q1 (b)	**4 / 8**	The answer is right — only x = 0, also written as the two-component zero vector, in line with the official solution's note that its size depends on n. The explanation, which the question asks for in bold, gives the wrong reason. The rank falling short of the number of rows is a fact about the column space: it is why (1, 1, 1) has no solution, and it says nothing about the nullspace — a 3 by 4 matrix of rank 2 has r < m and a two-dimensional nullspace. The reason is the uniqueness given in the question: if Ax = 0 had a nonzero solution, adding it to the solution for (0, 1, 0) would give a second one. Part (a)'s extra columns would create exactly such a solution, so the two parts also conflict.
Q1 (c)	**8 / 8**	Both examples are correct. The single column (0, 1, 0) works directly. The 3 by 2 matrix with columns (0, 1, 0) and (1, 0, 2) has independent columns; it reaches (0, 1, 0) only with x = (1, 0), and reaching (1, 1, 1) would need the second component of x to equal both one and one half.
Question 1 subtotal	**16 / 24**	

Part	Score	Grader's comment
Q2 (a)	**4.5 / 8**	The computed inverse is correct, found by Gauss–Jordan elimination once A was known. The first instruction is not answered anywhere: $A^{-1} = E_{23}E_{31}E_{21}$, the three operations multiplied in the order applied, with the last on the left. The working applies the three operations to A and sets the result equal to the identity — the relation this product comes from — but the product itself is never written, and it carries about three of the eight points. Half a point for an intermediate line: in the final augmented matrix the second row reads −4, 1, −1, where subtracting row 3 from row 2 gives −1, 1, −1. The stated answer uses the correct −1, so this reads as transcription rather than method.
Q2 (b)	**8 / 8**	Correct. Found by applying the three operations to a matrix of unknown entries, setting the result equal to the identity and solving the nine resulting equations, all solved correctly. Undoing the operations in reverse order on the identity is shorter, but the route taken is valid.
Q2 (c)	**8 / 8**	Correct, with the third operation rightly left out. Subtracting row 3 from row 2 works upward, so it belongs to the step from U to the identity, not to L; only the two downward multipliers, 4 and 3, sit below the diagonal.
Question 2 subtotal	**20.5 / 24**	

Part	Score	Grader's comment
Q3 (a)	**10 / 10**	Correct in both cases, with the elimination shown. When c = 3 the second column has no pivot, so the first and third columns of A form the basis; for every other c the first three columns do.

Part	Score	Grader's comment
Q3 (b)	**7 / 9**	The method is right in both cases — one free variable, x_4, when c is not 3, and two, x_2 and x_4, when c = 3 — and both special solutions for c = 3 are correct. For c other than 3 the vector is written (−2, 0, 1, 1), with +1 where the third entry should be −1; multiplied by A it gives (4, 4, 4), not zero, so it is not in the nullspace. The same vector appears correctly, as (−2, 0, −1, 1), in the c = 3 answer, so this reads as a sign slip, but the basis as written is wrong, and multiplying back by A would have caught it.
Q3 (c)	**9 / 9**	Correct. The particular solution (0, 1, 0, 0) works for every c, since the second column of A is exactly (1, c, 0). For c = 3 the answer also gives a second valid form starting from (1, 0, 0, 0), since the first column equals the right-hand side when c = 3; both forms describe the same set. The vector for c other than 3 repeats the sign slip from (b) and is not penalised a second time.
Question 3 subtotal	**26 / 28**	

Part	Score	Grader's comment
Q4 (a)	**8 / 8**	Correct: the nullspace has dimension at least two and at most five, since five columns with at most three pivots leave at least two free variables. The answer writes the dimension as R(N(A)); dimension is the right word, since rank belongs to a matrix rather than a subspace, but the meaning is clear.
Q4 (b)	**7 / 8**	Columns 1, 4 and 5 — the pivot columns — correctly identified as the ones that give the column space. The official solutions phrase this as forming a basis for it and weight it at seven of the eight points. The rest is missing: column 2 of A is four times column 1, and column 3 is the zero vector, both read directly from R, since row operations keep every relation among the columns.
Q4 (c)	**0 / 8**	Not attempted. Every 3 by 3 row-reduced echelon form is upper triangular, since each row’s pivot sits on or to the right of the diagonal with only zeros before it, and enough of them exist to reach every upper triangular matrix — six suffice, as the official solutions list. S is therefore the space of all upper triangular 3 by 3 matrices, which has dimension six.
Question 4 subtotal	**15 / 24**	

Report Card

Section	Earned	Available	Percent
Question 1 — Rank and the Number of Solutions	**16**	**24**	**67%**
Question 2 — Elimination Matrices, the Inverse, and A = LU	**20.5**	**24**	**85%**
Question 3 — Column Space, Nullspace, and Complete Solution	**26**	**28**	**93%**
Question 4 — Nullspace, Echelon Form, and Matrix Space	**15**	**24**	**63%**
Total	**77.5**	**100**	**78%**

Each part was assessed independently against the official solutions and scored on its own merits, without reference to the running total. The exam gives points per question only, so each question's points were split evenly across its parts: eight per part in Questions 1, 2 and 4, matching the official solutions' own weighting of 4(b), and ten, nine and nine in Question 3, where the extra point goes to part (a), whose elimination the other two parts build on. Full marks were awarded wherever the work is correct and complete, and where the student's method departs from the official solution but is valid it earns full credit. Deductions were taken only for an error on the page or for something a question asks for that the answer does not supply; every deduction is itemised above with the specific gap named, and an error repeated in a later part was penalised only where it first appears. The handwritten submission was read from rendered page images, and every matrix, vector and example on the page was checked by multiplying it out before scoring. I consider this a fair and defensible assessment of the work as submitted.

Claude (Opus 5.5) — Anthropic
Grader, at the student's request
Graded 11 October 2026

MIT 18.06SC, Fall 2011 — Unit 1 Exam. Assessed against the official solutions. Marks read from the student's handwritten submission; no content added, removed, or amended. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 18.06SC Linear Algebra, Fall 2011.